

CAPSTONE DESIGN PROJECT

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ABSTRACT

Multi-camera surveillance produces more footage than operators can review. This system detects, recognizes, and tracks faces across cameras with no enrolment. A fine-tuned YOLOv11-nano detector and an ArcFace recognizer run a CPU-only pipeline, resolving each person to one anonymous identity via a central Qdrant gallery. A self-trained ViT fragmented across pose, justifying the chosen pose-robust CNN.

PROBLEM STATEMENT & OBJECTIVES

PROBLEM

One operator cannot watch many feeds or cross-reference faces over time. The system must detect every face, assign a stable identity, and answer "when and where did this person appear?" — across 2+ cameras, in real time on CPU, no enrolment.

OBJECTIVES

- Detect faces in real time across 2+ cameras on CPU (YOLOv11-nano).
- Match each person to one identity across cameras and pose, without enrolment (ArcFace + Qdrant).
- Persist identities/timelines; serve a live dashboard (timeline, search-by-photo).

DESIGN CONSTRAINTS

ECONOMIC

Open-source only; runs on existing PCs — no GPU purchase

ENVIRONMENTAL

CPU inference + on-demand preview encoding limit energy use

SAFETY

No physical actuation; risk is informational (privacy)

ETHICAL

Anonymous-by-default IDs; demographic bias acknowledged

SUSTAINABILITY

Modular, local-only design; low-cost deployment

SOCIAL

Privacy-by-design; labels optional, never required

PROPOSED SOLUTION

TIER 1 · CAMERA SOURCES

Webcam

MJPEG · publisher.py

IP Camera

RTSP URL

Video File

any OpenCV source

↓ FRAMES

TIER 2 · SERVER (SINGLE MACHINE)

Supervisor

1 worker / camera

Pipeline Core

detect → track → align → embed

FastAPI

central identity matching

Reaper

stale → consolidate

Qdrant

512-d vectors · cosine

SQLite

Person · Camera · Sighting

↓ HTTP POLLING ~1–2 S · REST + MJPEG

TIER 3 · BROWSER

Next.js Dashboard

Live · People · Person timeline · Cameras · Search

Figure 1: Three-tier architecture — camera sources → single-machine server → browser dashboard.

METHODOLOGY

Capture

frame

YOLO

detect

conf ≥ 0.5

IoU track

keep id

Quality

gate

≥ 90px · sharp
≥ 60

YuNet

align

5 lmk →
112×112

ArcFace

embed

512-d · L2-
norm

Match ·

Qdrant

cos ≥ τ →
person

Sighting

open · beat ·
end

Figure 2: Per-frame pipeline — detect → track → align → embed → central match.

KEY RESULTS

0.873

Detector precision · WIDER FACE

99.5%

Recognition · LFW verification

5–15 ms

Per-face embedding · CPU

Cross-pose cosine similarity — same person, across pose

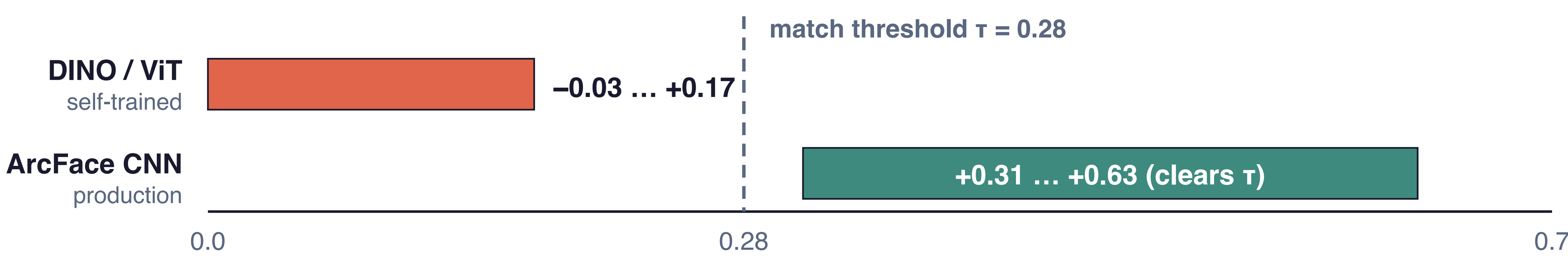


Figure 3: Cross-pose similarity — ArcFace clears the match threshold $\tau = 0.28$; the self-trained ViT collapses (≈ 0) and fragments identities.

ETHICAL & SOCIETAL IMPACT

SAFETY & HEALTH

No physical-actuation risk; the principal risk is informational (privacy), mitigated by local-only storage.

ENVIRONMENTAL

CPU inference and on-demand preview encoding limit energy use versus always-on GPU processing.

ETHICAL

Anonymous person_NNN IDs, no names required; biometric data stays local; inherited demographic bias acknowledged.

SOCIETAL BENEFIT

Security monitoring, access analytics, and missing-person search — balanced against surveillance and privacy risk.

CONCLUSIONS & FUTURE WORK

- 1 Detection met: YOLOv11-nano detector (precision 0.873) runs in real time on CPU across multiple cameras.
- 2 Recognition met: pose-robust ArcFace CNN (+0.31–0.63 cosine) holds one identity across pose; a self-trained ViT (≈ 0) fragmented.
- 3 Future work: GPU / batched model server for many HD cameras, server-push (SSE / WebSocket) UI, and stronger tracking (ByteTrack / DeepSORT).

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